A1

$$\begin{cases} x+2y+3z=1 \\ x+\alpha y+z=1 \\ 2x+3y+4z=2 \end{cases}$$

Ebatzi $\alpha=1$, $\alpha=2$ direnean.

$$A' = \left(\underbrace{\begin{matrix} 1 & 2 & 3 \\ 1 & \alpha & 1 \\ 2 & 3 & 4 \end{matrix}}_{A} \;\middle|\; \begin{matrix} 1 \\ 1 \\ 2 \end{matrix}\right)$$

$$|A| = 4\alpha + 4 + 9 - 6\alpha - 8 - 3 = -2\alpha + 2$$

$$|A| = 0$$

$$-2\alpha + 2 = 0 \qquad \boxed{\alpha = 1}$$

Sistema eztabaidatzeko Rouchéren teorema erabiliko dugu. Horretarako A eta zabalduaren heina aztertzen da. Heinak berdinak badira bateragarria izango da, n-ren berdina izan ezkero determinatua. Bestela indeterminatua. Heinak desberdinak badira BATERAEZINA

Kasuak $\begin{cases} \alpha \neq 1 \\ \alpha = 1 \end{cases}$

- $\alpha \neq 1$ denean

$|A| \neq 0 \rightarrow \text{hein}(A) = 3 = \text{hein}(A') = n$ $\boxed{\text{SIST. BATERAG. DETERMIN.}}$

- $\alpha = 1$ denean

$$|A| = 0.$$

$$\left(\begin{matrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{matrix} \;\middle|\; \begin{matrix} 1 \\ 1 \\ 2 \end{matrix}\right)$$

$$\left.\begin{matrix} |A| = 0. \\ \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = -1 \neq 0 \end{matrix}\right\} \text{hein}(A) = 2$$

$$\begin{vmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 2 \end{vmatrix} = 0 \quad \text{hein}(A') = 2$$

$\text{hein}(A) = \text{hein}(A') = 2 < n$ $\boxed{\text{SIST. BATERAG. INDETERM.}}$

Ebatzi:

$\alpha = 2$ denean SIST. BAT. DETERM. $\longrightarrow$ Cramer

$$A' = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 1 & 2 & 1 & 1 \\ 2 & 3 & 4 & 2 \end{array}\right) \qquad |A| = -2.$$

$$x = \frac{|A_x|}{|A|} = \frac{\begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 1 \\ 2 & 3 & 4 \end{vmatrix}}{-2} = \frac{-2}{-2} = 1$$

$$y = \frac{|A_y|}{|A|} = \frac{\begin{vmatrix} 1 & 1 & 3 \\ 1 & 1 & 1 \\ 2 & 2 & 4 \end{vmatrix}}{-2} = 0.$$

$$z = \frac{|A_z|}{|A|} = \frac{\begin{vmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 2 & 3 & 2 \end{vmatrix}}{-2} = 0.$$

$$\boxed{x = 1, \quad y = 0, \quad z = 0}$$

$\alpha = 1$ denean $\longrightarrow$ SIST. BAT. INDETER.

Batezgabetasuna det bihurtzeko eta Cramer erabiltzeko bat parametro bihurtu dugu eta ezezagun bi ekuazioak geratzen dira.

$z = \lambda$.

$$\begin{cases} x + 2y = 1 - 3\lambda \\ x + y = 1 - \lambda \end{cases} \Rightarrow \left(\begin{array}{cc|c} 1 & 2 & 1-3\lambda \\ 1 & 1 & 1-\lambda \end{array}\right)$$

$|B| = -1$

$$x = \frac{\begin{vmatrix} 1-3\lambda & 2 \\ 1-\lambda & 1 \end{vmatrix}}{-1} = \frac{1-3\lambda-2+2\lambda}{-1} = \frac{-\lambda-1}{-1} = \underline{\underline{\lambda+1}}$$

$$y = \frac{\begin{bmatrix} 1 & 1-3\lambda \\ 1 & 1-\lambda \end{bmatrix}}{-1} =$$

$$= \frac{\cancel{1}-\lambda-\cancel{1}+3\lambda}{-1} = \frac{2\lambda}{-1} = -2\lambda .$$

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$$\boxed{\begin{array}{l} x = \lambda+1 \\ y = -2\lambda \\ z = \lambda \end{array}}$$

B1)

$$A = \begin{pmatrix} \alpha & 0 & \alpha & 0 \\ 3 & \alpha & 0 & \alpha \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

A 3x4 dimentsioko matrizeen heina, asko jota hiru izan daiteke. Heina linealki indep. diren lerroen kopurua da. Heina kalkulatzeko matrizeen orden altuko minorra $\neq 0$ kalkulatzen dau.

Heina altuena 3 izango da, 3. ordeneko minorra $\neq 0$ doneean.

$$\begin{vmatrix} \alpha & 0 & \alpha \\ 3 & \alpha & 0 \\ 0 & 1 & -1 \end{vmatrix} = -\alpha^2 + 3\alpha$$

$-\alpha^2 + 3\alpha = 0$

$\alpha(-\alpha + 3) = 0$

$\alpha = 0$

$\alpha = 3$.

Kasuak aztertzen dira

- $\alpha \neq 0$ eta $\alpha \neq 3$ deneean heina$(A) = 3$.
- $\alpha = 3$
- $\alpha = 0$

$\alpha = 0$

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

lerro bat nulua dauet heina$(A) \leq 2$

$\begin{vmatrix} 3 & 0 \\ 0 & 1 \end{vmatrix} = 3 \neq 0$ heina$(A) = 2$

$\underline{\alpha = 3}$

$$\begin{pmatrix} 3 & 0 & 3 & 0 \\ 3 & 3 & 0 & 3 \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

$$\begin{vmatrix} 3 & 0 & 3 \\ 3 & 3 & 0 \\ 0 & 1 & -1 \end{vmatrix} = 0.$$

$$\begin{vmatrix} 3 & 0 & 0 \\ 3 & 3 & 3 \\ 0 & 1 & 2 \end{vmatrix} = 18 - 9 = 9 \neq 0 \quad \text{heiu}(A) = 3.$$

Oudbrist:

$$\begin{cases} \alpha = 0 & \text{deueon heiu}(A) = 2 \\ a \neq 0 & \text{deueon heiu}(A) = 3 \end{cases}$$